PRACTICAL PENETRATION TESTING EXERCISE

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**Topic:** *Network Reconnaissance Using Nmap*
**Date:** *15/07/2026*

**Introduction:**

This report presents the results of a network security scan conducted on the target system. The purpose of the assessment was to identify active hosts, open ports, running services, and potential security risks that may affect the security of the system.

**Summary:** This report presents the results of a network security scan conducted on a target system. The purpose of the assessment is to identify open ports, running services, and potential security risks that may exist on the system.

**Methodology**

**Tools Used**

- Nmap

**Commands Executed**

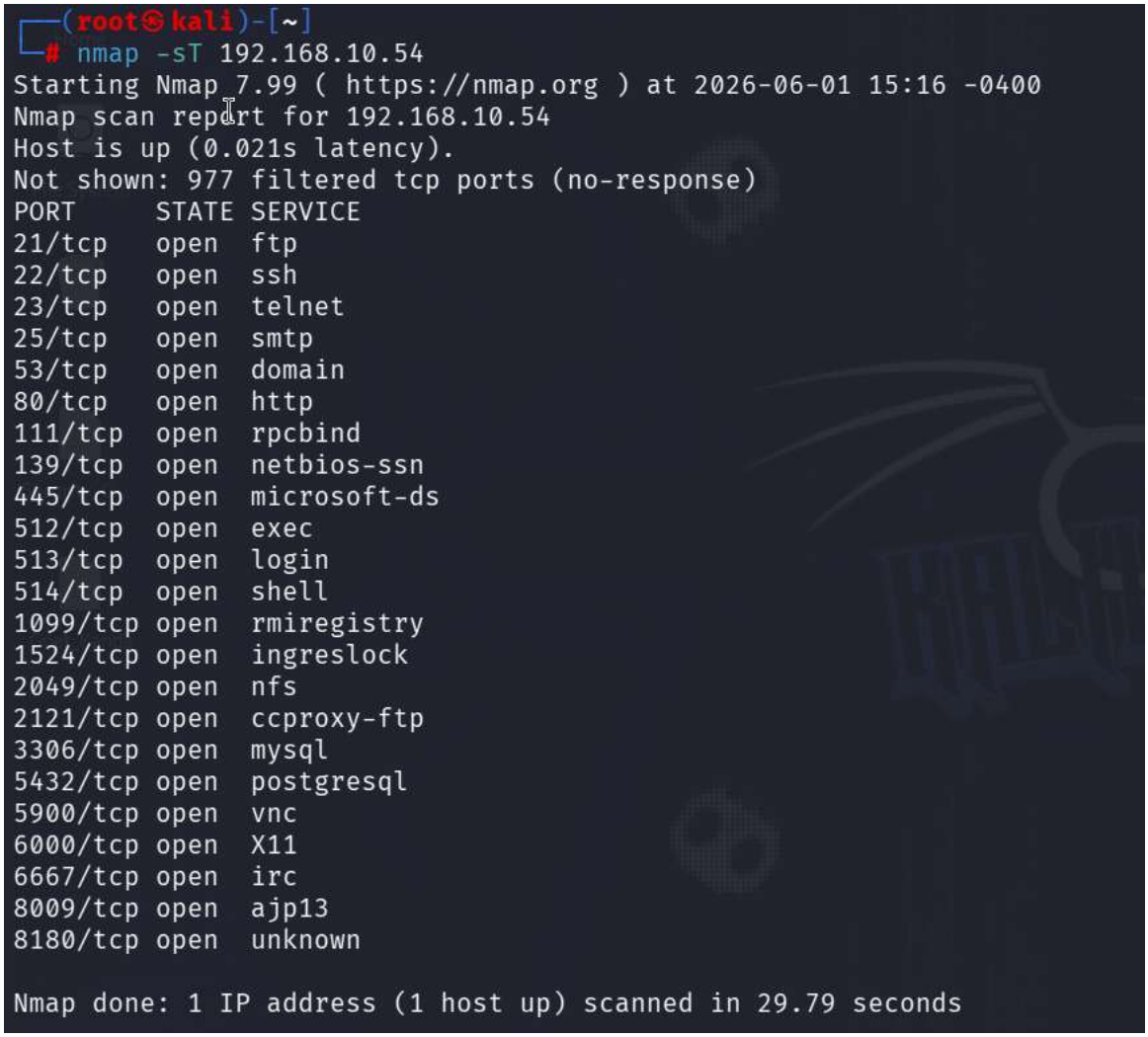
1. nmap -sn 192.168.10.54
2. nmap -p- -sV 192.168.10.54
3. nmap -sT 192.168.10.54
4. nmap -A 192.168.10.54

**Task 1:** After scanning and performing a ping scan of the **targeted IP**, the screenshot below **confirms** that the host is **online**.

**A successful host discovery**



**Task 2:** After performing a full TCP scan, the screenshot below shows the results for the open ports, service names.



**Task 3:** This screenshot below shows in detailed the detected service versions and the detected operating system.

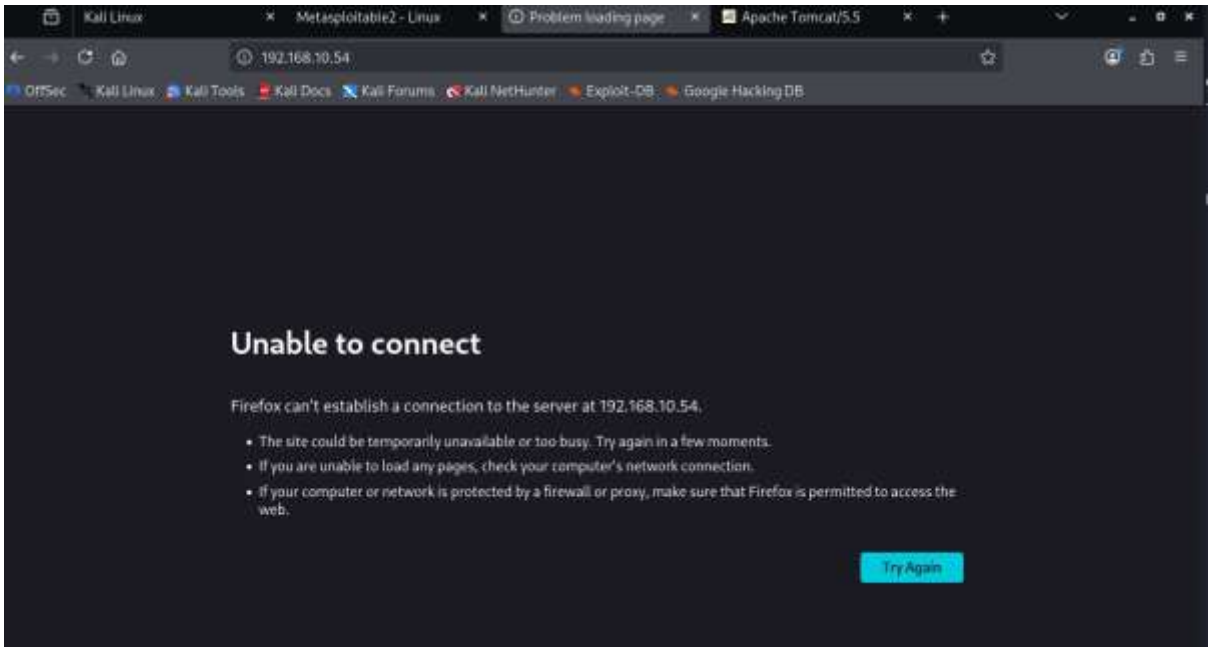



**Task 4:**

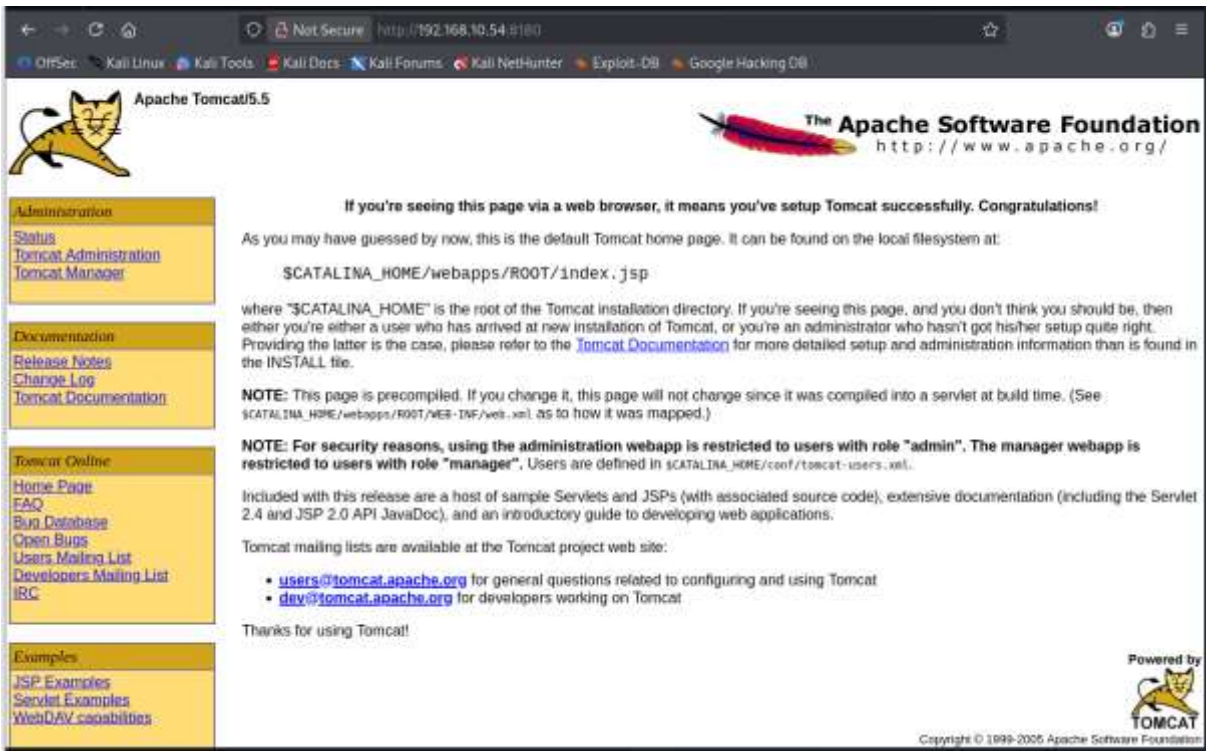
- **HTTP =** http://192.168.10.54



- **HTTP 80 =** http://192.168.10.54:8180




- **HTTPS =** https://192.168.10.54



## Task 5:

1. **How many open ports were discovered?**
   23 ports where discovered

2. **Which services are running?**
   All services are running

3. **Are any services potentially insecure?**
   Yes, some services are potentially insecure. Telnet, FTP, Exec, SMB, VNC and Database services.

4. **What security risks can you infer from exposed services?**
   It has many exposed services which an attacker can attempt an unauthorized access through insecure services such as Telnet, FTP and VNC. The presence of many open ports suggests that the system is vulnerable to reconnaissance and unauthorized access if security control is weak and not strong

## Screenshots

Screenshots of the scan results have been included in the report. Please refer to the attached images for the detailed Nmap output.

## Analysis

The scan indicates that the target system is running numerous network services, resulting in a large attack surface. Several services, including Telnet (23), FTP (21), RExec (512), RLogin (513), and RSH (514), are legacy protocols that transmit data without encryption and are considered insecure in modern environments.

In addition, database services such as MySQL (3306) and PostgreSQL (5432) are accessible, and file-sharing services such as NFS (2049) and SMB (445) are exposed. Remote access services including SSH (22) and VNC (5900) should be properly configured with strong authentication to reduce the risk of unauthorized access.

## Conclusion

The network scan reveals that the target system exposes a significant number of open ports and services, increasing its potential attack surface. Several legacy and high-risk services are enabled and could be exploited if left unsecured. It is recommended to disable unnecessary services, secure remote access protocols, restrict access using firewalls, apply current security patches, and continuously monitor the system to reduce security risks.